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The plasma proteome of an Ebola virus disease survivor reveals longitudinal changes in coagulation and innate immune pathways in the absence of therapy

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Abstract

Recurrent Ebola virus disease (EVD) outbreaks with high mortality emphasize the need for effective therapies administered after the onset of critical illness. Dysregulated coagulation and innate immune responses during severe EVD are an appealing therapeutic target. Here, we report longitudinal plasma proteome data to complement previously published viral kinetics and clinical course of a critically ill EVD patient cared for in the Special Clinical Studies Unit at the National Institutes of Health Clinical Center following randomization to the placebo arm of a clinical trial. We sampled blood during severe illness and recovery and measured 1297 human proteins' abundance by SomaScan relative to the first blood draw day and a healthy control cohort. We observed a broad reduction in total plasma proteome abundance on D12-D13 of illness coincident with cessation of Ebola virus replication in peripheral blood leukocytes. Targeted evaluation of proteins related to clot formation and breakdown and activation and function of platelets, monocytes, and neutrophils provided detailed insight into the identity, abundance, and timing of soluble mediators associated with critical illness onset, progression, recovery, and long-term sequelae. These observations can inform development of effective host-directed therapies.

Introduction

In March 2015, a 34-year-old male American health care worker was infected with Ebola virus (EBOV) in Sierra Leone. He was evacuated to the National Institutes of Health (NIH) Clinical Center in Bethesda, MD where he was enrolled in the multi-national PREVAIL II clinical trial (1). He was randomly assigned to receive the current standard of supportive care without an experimental therapy. The patient was admitted on day 7 (D7) of clinical illness and developed severe meningoencephalitis with sequential multi-organ failure (2). Cardiac magnetic resonance imaging performed early during recovery revealed myocarditis (3). Evaluation of peripheral blood leukocytes' transcriptional responses indicated a dramatic shift at D12 to D13 of illness coincident with EBOV nucleoprotein (NP) messenger (m)RNA levels, a surrogate for viral replication, becoming undetectable (4). During recovery, viral RNA remained detectable in the semen through D110 (5) and unbiased evaluation of his antibody repertoire against the complete EBOV proteome revealed robust EBOV-specific IgM responses through D361, consistent with ongoing antigen presentation (6).

There remains a shortage of information about the longitudinal dynamics of soluble immune responses during Ebola virus disease (EVD). Here, we applied bottom-up analyses to characterize plasma protein responses during acute illness and recovery using the 1.3K SomaScan v3 assay, an aptamer-based assay that measures 1297 human proteins' abundance with high sensitivity and specificity over eight orders of magnitude (i.e., femto- to micro-molar range) with remarkably low technical variability (7-10). 47% of measured proteins are secreted, 28% extracellular domains, and 25% intracellular belonging to broad biological subgroups including receptors, kinases, cytokines, proteases, growth factors, protease inhibitors, hormones, and structural proteins. We compared this patients' longitudinal analyte abundance values to his

baseline values on NIH admission and to pooled plasma of 21 healthy control subjects. By integrating circulating protein data into the existing high-resolution clinical and molecular picture of this severe EVD patient, we provide new insights into the identity, timing, and abundance of plasma proteins associated with disordered coagulation and early exuberant and late persistent immune responses contributing to EVD progression and resolution.

Results

Plasma proteome broadly shifts as EBOV stops replicating in circulating leukocytes

We previously reported (4) Ebola virus (EBOV) replication via detection of nucleoprotein (NP) mRNA in peripheral blood leukocytes (PBLs) on day 7 (D7) through D11 of clinical illness with a peak on D9 (**Figure 1A**). Active viral replication was not detectable on D12 or later. EBOV glycoprotein (GP) viral (v)RNA was detectable in serum by qRT-PCR as late as D22 (2).

In this study, we assessed the corresponding host proteome at the same time points as Ebola NP and GP quantification. We scanned the plasma with a set of 1297 human protein SOMAmers (**Suppl. Table 1**) from daily and intermittent samples taken during acute illness, recovery, and convalescence up to one year (D361) of clinical illness. Longitudinal profiles for individual proteins are shown in **Suppl. Figure 1**. We first compared the abundance of the EVD patient's plasma proteins to D7 protein levels obtained on the day of admission (**Figure 1B**).

The longitudinal profiles of individual proteins clustered into 4 distinct patterns (**Suppl. Table 1**). The majority of the proteins belonged to Cluster 1 (blue; **Figure 1C**) that exhibited a pattern of elevated abundance on D8 through D11, followed by a sharp decline in abundance

below D7 levels that persisted through D361. Cluster 2 (purple; **Figure 1D**) similarly contained proteins with elevated abundance through D11, which subsequently returned to and persisted near D7 levels. Cluster 3 (orange; **Figure 1E**) included proteins with persistent abundance near or above D7 levels through D361. Cluster 4 (brown; **Figure 1F**) included proteins that had persistently lower abundance through D361 when compared to D7. **Suppl. Figures 2A-B** show distributions of fold change profiles across the measured proteome by day of illness (relative to D7), which emphasize a marked transition on D12, which separates an early phase (D8-D11) where most proteins are detected significantly above D7 levels, and a late phase (D13 onwards) characterized by most proteins detected significantly below D7 levels. The longitudinal fold change profiles relative to D7, along with their associated p-values, are provided in **Suppl. Table 2**.

We also compared the abundance of the 1297 proteins in our EVD patient to the abundance in a healthy control cohort. A major transition point in the patient's plasma proteome compared to healthy controls occurred between D11 and D13. Between these days, abundance of 26.7% of proteins decreased to healthy control levels or below (**Suppl. Figures 2C-D**). There was a larger percentage of proteins with higher abundance compared to healthy controls than was observed when comparing to D7 levels. The percentage of proteins with higher abundance peaked on D10 when 68.7% of proteins had an abundance 3-fold or greater than healthy controls compared to only 2.2% of proteins which had an abundance 3-fold or greater above D7 levels. On D10, the abundance of 129 proteins (11.9%) were at least 50-fold higher than healthy controls with the highest increase being a 197-fold change (IFN- λ 2). Increased abundance was diminished on D12 when only 43.9% were at or above 3-fold higher than healthy controls. This decrease in proteins with higher abundance progressed further on D13 when only 13.9% of

proteins were 3-fold higher. The longitudinal fold change profiles relative to healthy control samples, along with their associated p-values, are provided in **Suppl. Table 3**.

Complex interplay of prothrombotic proteins, profibrinolytic proteins, and their inhibitors orchestrate coagulopathy and hemostasis

Clinical labs including a D-dimer >20ug/ml and an activated partial thromboplastin time (aPTT) of 74 seconds, along with an ISTH (International Society of Thrombosis and Hemostasis) DIC (disseminated intravascular coagulation) score > 5, previously confirmed that the patient had florid DIC by D7 of clinical illness (**Figure 2A**) (2). The DIC improved gradually after suppression of viral replication on D12. To gain insight into how the patient's coagulopathy changed over time through the lens of the plasma protein abundance, we curated a group of coagulation or fibrinolysis factors and inhibitors. We also took this approach for platelet-, monocyte-, neutrophil-, and complement-related proteins later in this report. A glossary containing the individual analytes selected for each category can be found in **Suppl. Table 4**.

The coagulation-related proteins were generally upregulated through D12 when compared to D7 (**Suppl. Figure 3B**). After D12, nearly all the proteins in the group belonged to either Cluster 1 (blue) with abundance dropping below D7 levels from D12 through D361, or to Cluster 3 (orange) with abundance remaining above D7 levels through D361.

On an individual analyte level, abundance of tissue factor (TF), the extrinsic coagulation initiator, surged above D7 levels through D11 (**Suppl. Figure 3B**) and above healthy controls through at least D28 (**Figure 2B**). Thrombin and coagulation factor V, additional clotting factors, were also at lower levels when compared to healthy controls through D9, suggesting these were being consumed during DIC. Abundance of thrombin-activatable fibrinolysis inhibitor (TAFI),

an inhibitor of fibrinolysis by preventing plasminogen binding to fibrin (11), was below healthy control levels on D7 before gradually returning to healthy control levels by D10. Simultaneously, abundance of fibrinolytic plasmin was increased during D8-D11 compared to D7 and healthy controls. Other pro-fibrinolytic enzymes such as tissue plasminogen activator (tPA) and urokinase-type plasminogen activator (uPA) were also elevated early compared to D7 and healthy control levels. Low levels of prekallikrein were yet another indicator of consumption via overactive fibrinolytic processes.

When EBOV replication ceased on D12, TF, pro-coagulation factor VII, clotting inhibitor activated protein C, and tPA decreased, while thrombin and TAFI returned to healthy control levels. At the same time, an abundance of plasminogen activator inhibitor-1 (PAI-1), a potent inhibitor of fibrinolysis, decreased from above D7 and healthy controls levels to below D7 and healthy control levels, indicating a brake was removed from clot breakdown on D12-14.

Pro-coagulation factors, including both TF and factor VII, were found above healthy control levels again from D18 through at least D28. Increased abundance of fibrinolytic factors such as uPA, tPA and plasmin along with another decrease in PAI-1 from D22-28 mirrored this increased trend in both enhanced clot formation and breakdown during recovery. In contrast, the abundance of TAFI was elevated above healthy control levels between D18-28.

When compared to healthy controls, there was persistent elevation of tPA, uPA, and tissue factor pathway inhibitor (TFPI) suggesting ongoing low-level clot formation and disruption during recovery, though this is unlikely to be clinically significant given normalization of D-dimer by D65 and D110 of clinical illness. By D361, TF and plasmin returned to near healthy control levels, but prekallikrein, factor V, factor X, and TFPI remained elevated indicating chronic impacts on some aspects of coagulation pathways.

Platelet-related proteins reveal sharp recovery when replicating EBOV is eliminated but also a need for persisting prothrombotic signals

In the setting of DIC, thrombocytopenia was observed between D4 and D15, with a nadir of 31K/ μ l on D11 followed by a period of reactive thrombocytosis on D20 (**Figure 2C**).

Analysis of selected platelet-related proteins yielded additional insights underlying platelet activity during these periods. During viremia, abundance of von Willebrand factor (vWF), a glycoprotein essential for platelet adhesion to endothelial wounds, was consistently elevated above healthy controls near D7 levels (**Figure 2D** and **Suppl. Figure 3D**). Platelet glycoprotein Ib alpha (GP1BA), a component of the heterodimer GPIb which mediates vWF binding and downstream platelet activation, was elevated above D7 and healthy control levels until D12. Abundance of thrombopoietin (Tpo), a liver-derived thrombopoiesis-stimulating hormone, was elevated during acute illness, tracking with the subsequent increase in circulating platelets. In contrast, CD39 (ectonucleoside triphosphate diphosphohydrolase-1) and platelet activating factor acetylhydrolase (PAFAH), enzymes which prevent platelet activation, were also elevated during viremia. Platelet factor 4 (PF-4), a chemokine produced by activated platelets that promotes coagulation and attracts leukocytes, was detected near healthy control levels during thrombocytopenia.

By D14, when EBOV was cleared from blood and platelet counts began recovering, GP1BA, CD39, PAFAH, and Tpo returned to near healthy control levels where they remained through D361. In contrast, vWF remained near D7 levels and above healthy control levels through D28 but returned to healthy control levels by D361. PF-4 levels became elevated compared to D7 and healthy controls through recovery and remained so at D361. Abundance of a

disintegrin and metalloproteinase with thrombospondin motifs 13 (ADAMTS13), a vWF-cleaving protease that protects against excessive thrombosis, decreased below healthy control levels during recovery but returned to near healthy control levels by D361.

Monocyte-related protein abundances reflect recovery from monocytopenia and immunoregulatory functions

Monocytes are primary targets for EBOV replication and cell lysis⁶, and correspondingly we observed below normal monocyte counts of <0.5K/ μ l on D7-8 of clinical illness (**Figure 3A**). Monocyte counts began to rise above normal as viral replication decreased. Abundance of selected monocyte-related plasma proteins were generally elevated above D7 levels until at least D13 (**Suppl. Figure 4B**) and healthy control levels until at least D14 (**Figure 3B**). Macrophage colony-stimulating factor 1 (M-CSF or CSF-1) and IL-34, cytokines that stimulate monocyte proliferation, differentiation, and survival, were elevated in the presence of EBOV replication and monocytopenia suggesting active processes to replenish monocyte numbers. CCL2 (MCP-1) and CCL7 (MCP-3), critical chemokines for the egress of monocytes from the bone marrow and recruitment to sites of infection, were also elevated on D8-11 compared to D7 and healthy controls. Levels of immunoregulatory factors including soluble macrophage scavenger receptor, soluble macrophage mannose receptor 1 (CD206), and matrix metalloproteinase (MMP)-12 were also elevated during D8-11 compared to D7 and above healthy controls through D13.

During monocytosis and recovery, M-CSF remained above healthy control levels, and macrophage-stimulating protein (MSP) levels increased above healthy controls through D28. MCP-1 and MCP-3 decreased closer to healthy control levels from D14-17 during monocytosis before rising again from D18-D25. MMP-12 and macrophage migration inhibitory factor (MIF)

both decreased below healthy control levels from D17-D22. By day 361, all the selected monocyte-related factors had returned to near healthy control levels except for MSP which remained elevated one year after illness onset.

Neutrophil-related proteins provide insights about specific timing of neutrophil migration and variations in granule release and persistence

Circulating neutrophils rose from 5.28 K/ μ l on D7 to a peak of 13.41 K/ μ l on D10 (**Figure 3C**). Neutrophilia persisted from D8 to D17, and we did not detect neutropenia on the days we sampled. Selected neutrophil-related proteins had a diversity of longitudinal kinetics with proteins falling into all four longitudinal pattern clusters relative to D7 (**Suppl. Figure 4D**).

Levels of promoters of neutrophil egress from the bone marrow, Gro- α (CXCL1) and GM-CSF, were elevated above healthy control levels through D12 (**Figure 3D**). Inversely, the level of G-CSF, a specific promoter of neutropoiesis, was detected below healthy control levels on D7-9. IL-17A, a pro-inflammatory cytokine that induces the production of G-CSF and the neutrophil chemokine, IL-8, was elevated compared to healthy controls until D25. ENA-78/CXCL5, IL-8, and IL-23, proteins that drive neutrophil migration and/or effector function activation were elevated above healthy control levels until D12, D22, and D28, respectively. Matrix metalloproteinase 9 (MMP9 or gelatinase B), an enzyme that facilitates neutrophil extravasation, was found to have fluctuating levels at or above healthy controls from D7-D16.

Primary neutrophil granule (azurophilic) antimicrobials neutrophil elastase, azurocidin, and cathepsin G were elevated above D7 levels until D11 and above healthy control levels until D13, D12, and D25, respectively. Other primary granule antimicrobials, like myeloperoxidase, proteinase-3, and bactericidal permeability-increasing protein (BPI), were not elevated when

compared to D7, but were considerably elevated when compared to healthy controls until D22, D22, and D11, respectively. Alpha-1-antitrypsin, an inhibitor of neutrophil elastase and proteinase-3, and alpha-1-antichymotrypsin, an inhibitor of cathepsin G and chymases, are both elevated through D28 compared to healthy controls.

Secondary granule antimicrobials had variable longitudinal patterns. Lactoferrin and peptidoglycan recognition protein 1 (PGRP-S) were detected at higher than healthy control levels until replicating virus was eliminated on D12. Secretory leukocyte protease inhibitor (SLPI; also known as antileukoprotease) gradually rises until plateauing around D12 when levels of the proteases it inhibits, neutrophil elastase and cathepsin G, begin to decline. SLPI remains elevated through D28 compared to healthy controls. Lysozyme has gram-positive antibacterial properties and is also present in tertiary granules and secretory fluids. Its circulating levels are downregulated during D8-9 and then rise and remain elevated above D7 and healthy control levels through D28. Lipocalin 2 (NGAL), is a siderophore-sequestering protein that prevents bacterial growth and is also considered an acute kidney injury (AKI) biomarker. Lipocalin 2 was initially slightly depleted compared to D7 levels; however, when compared to healthy controls, lipocalin abundance was elevated until D28 and trended with the patient's peak serum creatinine level of 6.2mg/dL (2).

On D361, none of the selected proteins that promote neutropoiesis, chemotaxis, or activation remained elevated above healthy controls. The same was true for circulating primary and secondary granule components. Gro-a, GM-CSF, elastase, ENA-78, and IL-8 were detected below healthy control levels. However, SLPI remained elevated one year after illness onset.

Classical, lectin, and alternative complement pathways recovered from a D7-8 bottleneck at different intervals

C1q and C1r are both essential factors for initiation of the classical complement cascade (12). C1q was detected below D7 and healthy control levels until D17 when it returned to healthy control levels, likely reflecting consumption (**Figure 4** and **Suppl. Figure 5**). Abundance of C1r, which requires C1q recognition of antigen-bound antibodies for activation, remained persistently elevated above healthy control levels from D7-361.

The antibody-independent lectin complement pathway is initiated when mannose-binding lectin (MBL), collectins, or ficolins recognize EBOV surface glycoproteins (13). Abundance of MBL is elevated compared to healthy controls from D7-11 in the presence of replicating virus. MBL abundance dropped below healthy control levels from D13-17, returned to healthy control levels from D18-28, and then reduced well below healthy control levels at D361. Collectin-11 (Collectin Kidney 1) and collectin-12 (COLEC12) circulated at near D7 levels and well above healthy control levels in the presence of replicating EBOV. Subsequently, Collectin-11 abundance dropped but remained above healthy controls until D28 before rising again on D361. Collectin-12 decreased to near healthy control levels by D17 and remained there. Ficolin-1 (FCN1), ficolin-2 (FCN2), and ficolin-3 were also all elevated above healthy controls in the presence of replicating EBOV. Subsequently, FCN1 and FCN2 abundance decreased but remained above healthy control levels through D25. Inversely, Ficolin-3 increased to its highest levels between D14-21 before returning to healthy control levels by D361. The lectin-pathogen binding activates MBL-associated serine proteases (MASPs). Similar to the lectins, abundance of MASP3 was elevated from D7-12 in the presence of EBOV. MASP3 abundance then decreased to near or below healthy control levels from D13-361.

Downstream of the initiation of the classical and lectin cascades, C4, a complement factor cleaved by C1 of the classical pathway and MASPs of the lectin pathway, was detected below healthy control levels until D16. C4b, a cleavage product of C4, had an abundance near healthy controls levels until D13-17 when its abundance decreased including a nadir on D14. Its abundance was highly above healthy controls on D361. In contrast, abundance of C2, also cleaved by C1, was increased through D11 compared to healthy controls. Subsequently, it reached a nadir on D14 before again rising above healthy controls on D18. C4 and C2 cleavage leads to cleavage of C3 and convergence with the lectin and alternative complement pathways (12).

C3 abundance was lower than the healthy controls through D12 after which it fluctuated near healthy controls. The abundance of C3a, a byproduct of C3 cleavage, was also below healthy controls until around D16 of infection. On the other hand, abundance of C3b, another byproduct of C3, a powerful opsonin, and a central mediator of the innate immune system, was below healthy controls D7-9 before increasing above healthy controls after D10-361. C3b is a component of the C5 convertase which is essential for cleaving C5 into C5a, a potent anaphylatoxin, and C5b, a foundational component of the terminal stage in the complement cascade, the membrane attack complex (MAC) (14). Abundance of C5 was below healthy control levels on D7 but rose well above healthy control on D9-11, peaking on D10. Abundance of C5a was also below healthy control levels on D7 but returned to healthy control levels by D9. Abundance of the C5b,6 complex, a critical intermediate in MAC assembly, was below healthy control levels on D7, but recovered to above healthy control levels on D8-11, reaching a sharp peak on D10. Subsequently, abundance of CD5b,6 complex decreased from its peak, but remained above healthy control levels through D28. C6 abundance was below healthy control

levels on D7-8 before restoring to healthy control levels. C7, C8, and C9, are the final components essential for forming the terminal pore complex and cytolysis of free enveloped EBOV virions or EBOV-infected cells (15). Abundance of all three factors was above healthy control levels from D7-11. After D11, C7 and C8 returned to healthy controls while C9 remained above healthy controls through D28.

In the antibody-independent alternative complement pathway, C3b binds directly to EBOV to initiate the cascade, and Factor B binds to C3b on EBOV surfaces. As mentioned above, C3b returned to healthy control levels on D9. Factor B remained near healthy control levels fluctuating below controls on D8 and above on D10-13. Factor D facilitates the activation of Factor B to form the active alternative pathway C3 convertase (C3bBb) and was elevated compared to D7 and healthy controls on D8-11 before returning to healthy control levels. The abundance of properdin, the only known stabilizer of the C3bBb convertase, was lower than healthy controls until D20 (16). Abundance of Factors H and I, negative regulators of the alternative pathway, were below healthy control levels on D7-9 before persisting near healthy control levels.

Abundance of other proteins involved in complement inhibition followed different longitudinal patterns. DAF circulated at higher levels than healthy controls from D8-28. Complement factor H-related protein 5 is elevated during D8-12 compared to healthy controls before abundance dropped. C1-esterase inhibitor's primary function is to inhibit the classical and lectin pathway, and abundance was consistently at or around baseline. TAFI levels were below healthy controls on D7 and subsequently increased to near or above healthy control levels. In addition, CD59, an inhibitor of MAC formation, was elevated during acute illness compared to D7 and healthy controls.

Discussion

We drew a series of conclusions from the overall plasma proteome presented relative to D7 and healthy controls (**Figure 1** and **Suppl. Figure 2**). First, the overall plasma proteome was more abundant on D7 than in healthy controls, indicating that the proteome had expanded prior to the time the patient was admitted. After this point, our statistical clustering analysis revealed that the proteins we analyzed had clustered into just four unique longitudinal protein abundance patterns, with the majority of the proteins clustering into one pattern with a marked peak between D8-11. Although NP mRNA peaked on D9, peak total protein abundance occurred on D10 suggesting that overall viral-mediated protein upregulation was delayed by up to 24 hours. The peak protein abundance on D10 aligned with the start of the patient's critical medical state (4) and the beginning of a profound rise in Sequential Organ Failure Assessment (SOFA) score (6). A marked decrease in abundance of the overall proteome occurred between D11 and D12 when compared to D7 and on D13 when compared to healthy controls. The inflection point coincided with the first days after cessation of EBOV replication in PBLs on D12 suggesting that EBOV viral replication had a meaningful impact on the plasma proteome of the critically ill patient. Comparing the overall proteome abundance of the patient and healthy controls also revealed a second smaller peak from about D18-D21. This peak coincided with reactive thrombocytosis and increased abundance of both coagulation and fibrinolysis factors, suggestive of an overshoot in the patient's compensatory pathway during recovery. SOFA score was decreasing from a D14 peak during this interval, but plaque reduction neutralization titers began to rise on D17 suggesting that onboarding of an effective adaptive antiviral immune response may have responsible for the changes during D18-D21 (6).

After EBOV replication was cleared and recovery began, some of the proteins we measured returned to healthy control levels, but many did not, as long as one year after illness onset. We interpreted these long-term differences as evidence of chronic protein dysregulation, another long-term consequence of EVD in line with previous observations of persisting EBOV (5) and anti-EBOV host immune responses (6).

Next, investigation of proteins related to coagulation revealed both prothrombotic and fibrinolytic processes occurring simultaneously during DIC and hemostasis. An increased abundance of TF during viremia and DIC compared to healthy controls and recovery time-points is consistent with prior studies that indicate increased abundance of TF on infected monocytes/macrophages and secreted microparticles (17), and is a primary contributor to consumptive coagulopathy in EVD. Although we do not provide direct evidence for this, the alignment of the rise and fall of TF with NP mRNA suggests that a major source of circulating TF is viral-bound TF that is derived by virions budding off of EBOV-infected monocytes (18).

Our data provided insights into the complex interplay of prothrombotic and fibrinolytic processes over time in EVD. While consumptive coagulopathy peaked on D6-7, plasminogen appeared to be consumed as it was activated into supraphysiologic levels of fibrinolytic plasmin. Persistently elevated plasmin, uPA, and tPA through at least D28 compared to healthy controls were consistent with active fibrinolysis and the patient's elevated D-dimer levels. D-dimer levels improved dramatically by D15 but were still elevated through D25, and the still elevated fibrinolytic proteins corresponded with this. Lower abundance of prekallikrein, required for plasmin activation, likely indicated it was being consumed, consistent with hyperactive fibrinolysis through much of this period. Elevated abundance of clotting inhibitors, activated protein C and TFPI, throughout the time course indicated that anti-coagulation mechanisms were

active in addition to fibrinolysis. Alpha-2-macroglobulin is a multifunctional inhibitor of both coagulation and fibrinolysis thought to protect against excessive clotting and degradation. Its abundance was lowest during D7-D9 before increasing, but it remained below healthy control levels until about D21, perhaps an indicator of the protracted coagulopathy.

The fibrinolysis inhibitors TAFI and PAI-1 were consumed at different times. The abundance of TAFI decreased below healthy controls on D7-9, and PAI-1 decreased below healthy controls on D12-14, possibly suggesting complementary roles to promote clot stability at different stages of illness.

Platelets are required at the start of the coagulation cascade to plug wounds and provide a surface for formation of fibrin mesh (19). Here, platelets are consumed during DIC, but mediators of platelet-endothelial binding and platelet activation, vWF and GP1BA, are found in excess. GP1BA found in the soluble form results from cleavage during platelet activation so its highly elevated abundance from D7-12 corresponds with consumption of activated platelets. CD39 and PAFAH, regulatory enzymes that can prevent excess platelet activation, were also found in excess during peak clinical DIC.

D13 marks a clear inflection point for these platelet-related proteins as platelet levels begin to rise toward normal levels. Correspondingly, GP1BA, CD39, and PAFAH abundance returned to healthy control levels indicating a return to normal levels of platelet activation. Abundance of Tpo dropped to healthy control levels or below indicating that there was no longer an emergency need for megakaryocyte stimulation. Soluble thrombopoietin receptor also returned to healthy control levels after being highly abundant during DIC. The meaning of the soluble receptor abundance is less clear, but its high levels during thrombocytopenia may reflect a compensatory response to clear excess TPO from circulation (20).

Despite these signs of a return to steady state, persistent elevated levels of vWF compared to healthy controls suggested that there was ongoing endothelial damage through D28. Elevated plasma fibronectin and PF-4 on D25, D28, and D361 could be further evidence of longer-term structural damage to blood vessels (21), and platelet hyperactivity, respectively (22).

Analysis of selected monocyte-related proteins yielded additional insights regarding the underlying monocyte activity during clinical monocytopenia and viral replication. A burst of growth factors (CSF-1 and IL-34) and chemokines (CCL2 and CCL7) well above healthy control levels likely was critical to the subsequent increase in monocyte count. The abundance of each of these proteins did not return to healthy control levels until D28 suggesting a continued need for monocyte recovery, and likely tissue repair, through D25.

MMP-12 was elevated above healthy controls D7-13 after which it returned to normal or lower than normal levels. It is an enzyme primarily produced by activated macrophages and has a wide range of functions after tissue injury including extracellular matrix remodeling, promoting monocyte and neutrophil chemotaxis, and also regulating type 1 immune responses (23). Elevated abundance of sCD163 above healthy control levels from D7-28 is thought to correspond with monocyte activation because soluble forms of this scavenger receptor are shed from activated monocytes (24). In addition to being an established marker of macrophage activation, soluble macrophage scavenger receptors such as sCD163 are known to have a variety of functions including hemoglobin scavenging and a decoy receptor for cytokines (24). Elevated sCD163 levels have been associated with fatal outcomes in EVD (25) suggesting that this patient survived despite being on track for a fatal outcome when he was admitted for supportive care. Elevated abundance of macrophage mannose receptor (sCD206) persisted from D7 through D28

but returned to healthy control levels by D361. Its elevation is also a biomarker for inflammation, and it promotes a pro-inflammatory macrophage phenotype (26).

Macrophage stimulating protein (MSP) is highly elevated after D17 and on D361. There are limited studies about its function, but several agree that MSP can negatively regulate nitric oxide production and instead promote alternative activation of macrophages (27, 28). This suggests that chronic high abundance of MSP may be supporting ongoing tissue repair through macrophage polarization.

Analysis of the selected neutrophil-associated proteins yielded further insights into the temporal dynamics of the innate immune response. Firstly, the elevated abundance of promoters of neutrophil egress from the bone marrow (Gro-a and GM-CSF) in the presence of replicating EBOV (through D11-12) correspond with a need for emergency granulopoiesis to replace mature neutrophils that followed EBOV out of the blood and into tissues (29). Potent neutrophil chemokines IL-8 and ENA-78 were highly elevated above healthy controls through D12 providing evidence that neutrophils were getting activated and trafficking into and out of the blood towards EBOV infection (30). Notably, abundance of IL-8 remained elevated (although less highly elevated) through D22 suggesting ongoing neutrophil chemotaxis until that time. IL-23 acts upstream of IL-17, and IL-17 acts upstream of IL-8 and ENA-78 (31), so elevated abundance of IL-23 and IL-17 until D28 and D25, respectively, were also consistent with neutrophil migratory activity through that stage of disease. IL-17 also acts upstream of G-CSF, a specific promoter of neutropoiesis that had an abundance below healthy controls on D7-9 and D16-22. Because IL-17 was present in excess throughout this period, the low levels of G-CSF likely reflect consumption and sustained neutropoiesis.

We closely analyzed some of the unique pre-formed proteins released from neutrophil granules into the plasma. Abundance of MMP-9, which aids neutrophil trafficking through extracellular matrix and can pathogenically contribute to vascular permeability (32, 33), was elevated above healthy controls on D7, D10-11, and D15-16 before returning to healthy control levels perhaps suggesting that most neutrophil extravasation out of the blood and into tissues ended by D17. Elevated abundance of potent antimicrobial primary and secondary granule proteins such as elastase, azurocidin, lactoferrin, and PGRP-S coincided with presence of replicating EBOV in the blood compartment. They did not persist in the blood despite the continued inflammatory environment likely due to potent requirements for primary and secondary granule release (34). Abundance of some of these proteins including cathepsin G, myeloperoxidase, proteinase-3, lipocalin-2, and lysozyme remained elevated for another week or more, however. Persistent elevation of these enzymes through at least D25 is associated with both antimicrobial benefits, but also ongoing inflammation and destruction of host tissues D25 due to the roles of these proteins in oxidative stress, immune cell chemotaxis, and vasculitis (30, 35). For example, renal function began to improve about D21, but elevated abundance of lipocalin-2, a marker of acute kidney injury (36), persisted until at least D28. Persisting elevation of inhibitors of these enzymes, alpha-1-antitrypsin, alpha-1-antichymotrypsin, and SLPI, through D28, further supports ongoing need for host protection. One year after illness onset, all of the selected neutrophil-related proteins returned to healthy control levels except for SLPI, indicating an ongoing need for protection of epithelial surfaces from proteolytic enzymes.

Analysis of the longitudinal abundances of individual complement-related proteins in the plasma provided insights into activities of the classical, lectin, and alternative complement pathways. At the start of the classical pathway, abundance of the essential activating protease

C1r was elevated above healthy controls from D7-361. Interferon-induced Signal Transducer and Activator of Transcription (STAT) activation likely boosted C1r expression which persistantly primed the classical complement pathway for a year after illness onset (37). However, C1q binding to immune complexes, pathogens, or apoptotic cells is required upstream of C1r activation (12). C1q abundance was below healthy controls from D7-17 before persisting at healthy control levels suggesting that C1q was consumed by binding to EBOV or immune complexes until abundance of EBOV sufficiently decreased. Like C1q, C4 appeared to be consumed during the same interval. C1q and C4 reached their nadirs on D11-12 suggesting that demand for the classical pathway peaked at this time, and that there the classical pathway was stuck in a bottleneck until about D17 when C1q levels returned to healthy control levels. The C4 cleavage byproduct, opsonin, and amplifier of the complement cascade, C4b, reached its nadir at D14 after several days of low C4 abundance. Notably, C4b was highly elevated on D361 compared to healthy controls indicating ongoing complement pathway amplification that could have been driven by the patient's persisting viral repertoire and/or microvascular damage.

We observed supraphysiologic levels of circulating lectin pathway initiators MBL, collectin-11, collectin-12, and ficolins 1, 2, and 3, from D7-11 in the presence of replicating virus. High serum concentrations of MBL in mice been shown to promote survival from otherwise fatal EBOV infection (38). Reduction of most of these proteins along with MASP3, a key regulator of the lectin pathway, towards healthy control abundance after D12 suggested that there was not marked ongoing activation of the lectin pathway after replicating EBOV was cleared from PBLs. Consumption of C4 and C4b on D13-17 was evidence that the lectin pathway also faced a bottleneck during this period. On D361, it is unclear why MBL abundance

was well below healthy controls. Inversely, collectin-11 abundance was elevated on D361 perhaps to compensate for diminished MBL.

We observed a few signs of demand for the alternative pathway, but no indications of a bottleneck after D8 when abundance of C3b, Factor B, and Factor D were present above healthy control levels. On D7-10, abundance of negative regulators Factor H and I was below healthy controls which could be due to consumption or viral evasion (39, 40). Despite neutrophilia, primarily neutrophil-sourced properdin abundance was lower than healthy controls through D18 indicating its consumption and a demand for stabilization of the alternative pathway's C3 convertase enzyme throughout this period (16).

Collectively, the individual complement factors that we examined provided evidence that all three pathways were highly activated and facing a consumptive bottleneck until D9. C3, the convergence point of the classical, lectin, and alternative complement pathways, appeared to be consumed in the presence of replicating EBOV from D7-12. Its nadir was at D7 with steady recovery after the time of admission. Low abundance of C3b and C5a from D7-8 indicated that efficient and sustained MAC formation was not occurring until after the patient began to receive supportive care. Inversely, C7, C8, and C9, the final components necessary for MAC formation, were found in high abundance from D7-12 perhaps suggesting that they were accumulating while upstream factors were getting consumed.

After D7-8, we found evidence that continued consumption of C1q, C3, and C4 for several more days was overcome by sufficient C3b (a cleavage product of C3, crucial opsonin, and trigger for complement activation and formation of the MAC), Factor B, and Factor D to create the C5 convertase and downstream MAC assembly via the alternative complement pathway.

Several factors also indicated longer-term complement activation. Abundance of C5a, an anaphylatoxin, and MAC assembly components C5, C5b,6, and C9 was elevated through D28. C3b, C1r, C4b, and Collectin 11 remained highly abundant compared to healthy controls from D10-361, suggesting that there was chronic uncontrolled complement activation for a year after symptom onset that can accompany ongoing inflammation and tissue damage. In part, EBOV persistence in immune protected sites may be contributing to the ongoing activation (5, 6).

While we conducted this study because of the strength of high-frequency sampling of a patient over time, the applications of our data to the general population may be limited by only studying a single patient. We selected a finite list of analytes to reveal novel time-specific features of coagulation and aspects of the innate immune response during EVD. We acknowledge that DIC, complement cascades, and other innate immune pathways are extremely complex, and our approach to draw conclusions from an incomplete list of involved proteins can introduce bias. We also recognize the limitations of relying on relative protein abundance data without functional assays. Consequently, we were careful to not overstate our conclusions and to focus on analytes that were essential to pathways to minimize bias.

To our knowledge, our results build on few prior studies of the plasma proteome after Ebola virus infection. One study used LC-MS/MS to report significant changes in the abundance of 41 of about 225 proteins collected over six days of a rapidly progressing lethal model of Ebola virus infection in nonhuman primates (41). Another study used LC-MS to compare 1000 host or viral proteins in the plasma of 12 humans with EVD (42). The study identified distinct protein signatures in survivors and fatalities, but longitudinal samples were not taken with high frequency over an extended disease course, and interferon beta 1a treatment impacted the host proteome.

Here, we were able to match high-frequency longitudinal proteome data with clinical and physiological data from an EVD survivor who did not receive an experimental therapy. This approach enabled us to advance understanding of the proteins responsible for coagulation and innate immune dysregulation at the center of both severe illness and recovery of patients with EVD. Two monoclonal antibody (mAb)-based therapies are FDA-approved for the treatment of EVD, given proven efficacy to reduce mortality, particularly when started early in the disease course (43). No proven effective therapies are available for Sudan virus disease or Bundibugio virus disease. If started later during EVD, however, when inflammation and coagulation predominate, mAb therapy is associated with refractory high mortality of 60-70%. In these patients, therapeutic approaches that target both pathogen and pathological host-responses are needed. Our data provide extensive longitudinal insight into putative cellular and molecular targets for such therapies.

Methods

Sex as a biological variable

Our study examined only a male individual because longitudinal plasma samples are not available for other Ebola virus disease survivors. The findings are expected to be relevant beyond the male sex.

Plasma irradiation

All clinical samples from the EBOV-infected patient were irradiated with 5 Mrad before removal from biosafety level-4 under supervision of NIH Institutional Biosafety and in accordance with

Division of Select Agents and Toxins at the Centers for Disease Control and Prevention. Pooled healthy control plasma was obtained from 21 healthy adult donors (11 female, 10 male) of median age 57 (Q1 = 37, Q3 = 61.5) years with the Trans-NIH Center for Human Immunology, Autoimmunity, and Inflammation (CHI) at the NIH; healthy control plasma samples were identically irradiated.

Proteome characterization

Proteomic analysis was performed on plasma collected daily from a single patient with EVD and intermittently during recovery up to 361 days of clinical illness. Data were obtained using the SomaScan Assay (SomaLogic, Inc.) at the NIH CHI. The 1.3K SomaScan Assay was run as previously reported (8). In brief, irradiated plasma samples were analyzed using the High-Throughput-System (HTS) 1.3K SomaScan Assay (SomaLogic; Boulder, CO), a sensitive and quantitative proteomic biomarker discovery platform that relies on DNA-modified aptamers (SOMAmers) designed to optimize high affinity, slow off-rate, and high specificity to target proteins (7). Twelve spike-in SomaScan reagent controls were included for hybridization normalization. Recombinant protein quality control and calibrator samples were used to correct for interplate variability following procedures previously developed for SomaScan bioinformatics (8, 44). It should be noticed that, as reported previously (8), 5 SOMAmers had to be removed following SomaScan's change log from December 2016 (SeqId 2795-23, 3590-8, 5071-3, 5118-74, 5073-30) and additional 8 SOMAmers were filtered out because they were not annotated as Organism="Human" or Type="Protein" (SeqId 5437-63, 5440-26, 5441-67, 5443-62, 2623-54, 2624-31, 2769-3, 4792-51). The final list consisting of 1297 SOMAmers targeting human proteins is provided in **Suppl. Table 1**.

Directed SOMAmer selection

Specific SOMAmers were selected *a priori* for directed analysis within the context of interest related to coagulopathy or innate immune responses to EVD, including: coagulation and fibrinolysis, platelets, monocytes/macrophages, neutrophils, and complement system. Selected proteins were catalogued with brief functionality annotations (**Suppl. Table 4**).

Clustering of longitudinal profiles

Protein abundance in relative fluorescence units (RFUs) was arranged in matrix form with 1297 rows (analytes) and 19 columns (timepoints). The cosine similarity (c) between each pair of longitudinal profiles (rows) was calculated (cosine function in R package *lsa* v.0.73.4) and their pairwise dissimilarity (d) was determined as $d = \sqrt{2(1-c)}$, which satisfies ultrametric properties (45). Based on this dissimilarity matrix, hierarchical clustering was performed using average linkage (*hclust* function in R base package *stats* v.4.5.1). A grid search approach was then implemented to generate sequential cuts of the hierarchical clustering dendrogram (*cutreeHybrid* function in R package *dynamicTreeCut* v.1.63.1) for different sets of parameters chosen for *deepSplit* (an integer in the 0-4 range that provides control over cluster splitting sensitivity) and the minimum cluster size (at 5% or 10% of the number of analytes). For each set of parameters, we determined the Pearson gamma coefficient (*cluster.stats* function in R package *fpc* v.2.2.13), which is a metric used to assess the quality of a clustering partition by measuring the correlation between data distances and cluster assignments. Higher values of the Pearson gamma coefficient indicate better clustering quality, signifying that distances between points in different clusters are consistently larger than those within the same cluster (46). From these

observations (**Suppl. Figure 6A**), we determined that the optimal solution was that of deepSplit=1 and minimum cluster size=5%, leading to four longitudinal patterns (**Suppl. Figure 6B**). With the exception of one analyte (Carbonic anhydrase 6), which was assigned to the noise cluster 0, the remaining 1296 analytes were classified into one of the four longitudinal patterns labeled as Clusters 1-4. Clustering results for all SOMAmers are provided in **Suppl. Table 1**. In addition, we determined the silhouette score for each SOMAmer (silhouette function in R package cluster v.2.1.8.1), which is a measure of how similar an object is to its own cluster (cohesion) compared to other clusters (separation). The silhouette score ranges from -1 to +1, where a high value indicates that the object is well matched to its own cluster and poorly matched to neighboring clusters. Using a silhouette threshold of 0.2, we find 1051 (81%) analytes assigned to Clusters 1-4 above the threshold (**Suppl. Table 1**). Heatmaps were generated using the heatmap.2 function in R package gplots v.3.2.0. Within each longitudinal pattern, analyte profiles were hierarchically clustered using its default function (i.e. hclust with complete linkage based on Euclidean distance). Data analysis and visualization were performed via custom scripts implemented in R v.4.5.1.

Clinical labs

White blood cell differential counts were performed at the NIH Clinical Center by the Department of Laboratory Medicine using standard methods. D-dimer, platelet counts, and activated partial thromboplastin time (aPTT) were measured using standard clinical laboratory methods.

Statistics

Wood (47) and Reed et al.(48) showed how to utilize the coefficient of variation (CV), defined as the standard deviation divided by the mean over technical replicate measurements, to perform rigorous statistical tests of significance. The probability that two replicate measurements differ by a fold-change equal or larger than FC is given by $p(FC) = 2 N_c \{-\log_e(FC)/\sqrt{2} \log_e(CV^2+1)\}$, where N_c is the cumulative standard normal distribution function and the fold-change is restricted to $FC \geq 1$, without loss of generality. For each protein measured in the SomaScan assay, we calculated the longitudinal FC profile from the Ebola patient's samples relative to day 7 (**Suppl. Table 2**) and relative to the mean protein abundance observed in identically irradiated pooled healthy control samples (**Suppl. Table 3**). To characterize the statistical significance of each of these FC values, we calculated the probability for the observed FC via Reed et al.'s formula, where CV was estimated from technical replicates of irradiated pooled healthy control samples. P-values from this analysis are provided in **Suppl. Tables 2-3**.

Study approval

Blood samples from the EVD patient (clinicaltrials.gov identifier NCT02363322) were obtained under clinical protocols approved by the National Institutes of Health (NIH) National Institute of Allergy and Infectious Diseases (NIAID) Institutional Review Board and were conducted in accordance with provisions of the Declaration of Helsinki and good clinical practice guidelines. Healthy volunteers samples were obtained via donation at the NIH Clinical Center and were not affiliated with patient-identifying information.

Data and software availability

To ensure the transparency and reproducibility of our results, we provide a dedicated Open Science Framework repository (<https://osf.io/mkaub>) to permanently store the processed datasets and software scripts needed to re-generate the results and figures presented in our study. Further inquiries can be directed to the corresponding authors.

[NOTE TO REVIEWERS: The Open Science Framework repository will be made public upon publication of this paper. For confidential, blind peer review, please use this anonymized link:

https://osf.io/mkaub/overview?view_only=dc8498a2a602401785b0c530d06d29f6]

Author contributions

Conceptualization, DSC and LJPV; Data Analysis, LJPV, JC, JA, DSC, AB; Resources, DSC and RTD; Data Curation, LJPV, JC, JA, DSC, MJRB, KMV; Bioinformatic and Statistical Analysis, Figure Preparation, Software and Data Repository, JC; Writing-Original Draft Preparation, LJPV, JC, MJRB, TW, KMV; Writing-Review & Editing, LJPV, KMV, JA, JC, TW, MJRB, DSC, AB, RTD.

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Figure Legends

Figure 1. Longitudinal plasma protein abundance relative to D7 .

A. Daily EBOV nucleoprotein mRNA isolated from peripheral blood mononuclear cells measured by strand-specific qRT-PCR, as previously reported (4). **B.** Heatmap showing overall plasma protein abundance. Each column represents the relative protein abundance compared to the protein abundance on D7 of clinical illness. Abundance of proteins shown in green were increased, proteins shown in red were decreased, and proteins in black were unchanged compared to protein abundance on D7. A clustering procedure applied to the longitudinal protein profiles identified four distinct clusters, shown by colors blue, purple, orange and brown on the leftmost column along the y-axis. **C-F.** Corresponding to each cluster, panels C-F show the median abundance value of all proteins within each cluster, respectively, with error bars representing the median absolute deviation. Panels B-F display results for 1051 (81%) analytes above the silhouette threshold of 0.2; detailed results for all 1297 SOMAmers are provided in **Suppl. Table 1.**

Figure 2. Longitudinal plasma abundances of selected coagulation and fibrinolytic factors and their inhibitors relative to healthy controls.

A. Clinical measures of coagulation: activated partial thromboplastin time (aPTT; purple) and D-dimer (gold). **B.** Heatmap showing abundance of selected proteins related to coagulopathic or fibrinolytic pathways in the plasma relative to healthy controls. **C.** Absolute platelet counts determined by complete blood count (CBC). **D.** Heatmap showing abundance of selected proteins expressed by or involved with platelet function in the plasma relative to healthy controls. Clinical measures on all days, with the exclusion of D65 and D110, were previously

reported (4). Abundance of proteins shown in green were increased, proteins shown in red were decreased, and proteins in black were unchanged compared to protein abundance of controls. Longitudinal profile clusters are indicated by the color-coded columns to the left of each heatmap (Cluster 1: blue; Cluster 2: purple; Cluster 3: orange). Asterisks indicate analytes clustering below the silhouette threshold of 0.2.

Figure 3. Longitudinal plasma abundances of selected monocyte- and neutrophil-related proteins relative to healthy controls.

A. Absolute monocyte counts determined by CBC. **B.** Heatmap showing abundance of selected proteins related to monocytes in the plasma relative to healthy controls. **C.** Absolute neutrophil counts determined by CBC. **D.** Heatmap showing abundance of selected proteins related to neutrophils in the plasma relative to healthy controls. Clinical measures on all days, with the exclusion of D65 and D110, were previously reported (4). Abundance of proteins shown in green were increased, proteins shown in red were decreased, and proteins in black were unchanged compared to protein abundance of healthy controls. Longitudinal profile clusters are indicated by the color-coded columns to the left of each heatmap (Cluster 1: blue; Cluster 2: purple; Cluster 3: orange; Cluster 4: brown). Asterisks indicate analytes clustering below the silhouette threshold of 0.2.

Figure 4. Longitudinal plasma abundances of selected complement-related proteins relative to healthy controls.

Heatmap showing abundance of selected proteins related to complement pathways in the plasma relative to healthy controls. Abundance of proteins shown in green were increased, proteins

shown in red were decreased, and proteins in black were unchanged compared to protein abundance of healthy controls. Longitudinal profile clusters are indicated by the color-coded column to the left of the heatmap (Cluster 1: blue; Cluster 2: purple; Cluster 3: orange). Asterisks indicate analytes clustering below the silhouette threshold of 0.2.

Supplementary Figure 1. Longitudinal profiles for 1297 proteins. For each protein, we show the Ebola patient's longitudinal profile (red) compared with the measured range of pooled healthy plasma controls (blue). Protein abundance is expressed in Relative Fluorescence Units (RFU). Proteins are identified with the Entrez gene symbol and (when different) the protein name, followed by "SeqId", the SOMAmer unique identifier, in parenthesis. See **Suppl. Table 1** for further details on each measured protein.

Supplementary Figure 2. Distributions of fold change profiles across the measured proteome by day of illness.

A. Profiles shown for each day are in a different color. Curves represent the fraction of proteins (y-axis) whose abundance relative to D7 is more extreme (farther than 1) than given fold change (FC) values (x-axis). Example: on D10 (yellow curve), approximately 90% of all proteins had abundances larger ($FC > 1$) than D7; likewise, about 50% of all proteins had abundances larger than two-fold ($FC > 2$) those of D7; however, the curve quickly drops, showing that only about 10% of proteins had an $FC > 2.5$ relative to D7 protein levels. A marked transition appears to occur on D12, which separates an early phase (D8-D11) where most proteins are detected significantly above D7 levels and a late phase (D13 onwards) characterized by most proteins detected significantly below D7 levels. **B.** Same as panel A, but showing an extended FC range

up to 10-fold above (FC=10) / below (FC=0.1) D7 protein levels. **C-D.** Similar to panels A-B, but using irradiated pooled plasma samples from 21 healthy controls as reference protein levels.

Supplementary Figure 3. Longitudinal plasma abundances of selected coagulation and fibrinolytic factors and their inhibitors relative to D7.

A. Clinical measures of coagulation: activated partial thromboplastin time (aPTT; purple) and D-dimer (gold). **B.** Heatmap showing abundance of selected proteins related to coagulopathic or fibrinolytic pathways in the plasma relative to D7. **C.** Absolute platelet counts determined by complete blood count (CBC). **D.** Heatmap showing abundance of selected proteins expressed by or involved with platelet function in the plasma relative to D7. Clinical measures on all days, with the exclusion of D65 and D110, were previously reported (4). Abundance of proteins shown in green were increased, proteins shown in red were decreased, and proteins in black were unchanged compared to protein abundance on D7. Longitudinal profile clusters are indicated by the color-coded columns to the left of each heatmap (Cluster 1: blue; Cluster 2: purple; Cluster 3: orange). Asterisks indicate analytes clustering below the silhouette threshold of 0.2.

Supplementary Figure 4. Longitudinal plasma abundances of selected monocyte- and neutrophil-related proteins relative to D7.

A. Absolute monocyte counts determined by CBC. **B.** Heatmap showing abundance of selected proteins related to monocytes in the plasma relative to D7. **C.** Absolute neutrophil counts determined by CBC. **D.** Heatmap showing abundance of selected proteins related to neutrophils in the plasma relative to D7. Clinical measures on all days, with the exclusion of D65 and D110, were previously reported (4). Abundance of proteins shown in green were increased, proteins

shown in red were decreased, and proteins in black were unchanged compared to protein abundance on D7. Longitudinal profile clusters are indicated by the color-coded columns to the left of each heatmap (Cluster 1: blue; Cluster 2: purple; Cluster 3: orange; Cluster 4: brown). Asterisks indicate analytes clustering below the silhouette threshold of 0.2.

Supplementary Figure 5. Longitudinal plasma abundances of selected complement-related proteins relative to D7.

Heatmap showing abundance of selected proteins related to complement pathways in the plasma relative to D7. Abundance of proteins shown in green were increased, proteins shown in red were decreased, and proteins in black were unchanged compared to protein abundance on D7. Longitudinal profile clusters are indicated by the color-coded column to the left of the heatmap (Cluster 1: blue; Cluster 2: purple; Cluster 3: orange). Asterisks indicate analytes clustering below the silhouette threshold of 0.2.

Supplementary Figure 6. Grid search approach to find the optimal clustering solution.

Longitudinal plasma protein abundance profiles (relative to D7) were hierarchically clustered, then sequential cuts of the resulting dendrogram were implemented using different values for the deepSplit parameter (integers in the 0-4 range) and minimum cluster size thresholds of 5% (65 analytes) or 10% (130 analytes). **A.** Pearson gamma coefficients were used to assess the quality of a clustering partition by measuring the correlation between data distances and cluster assignments. Higher values of the Pearson gamma coefficient indicate better clustering quality, signifying that distances between points in different clusters are consistently larger than those within the same cluster. **B.** Number of clusters obtained for each parameter choice.

Supplementary Table 1. List of all 1297 analytes measured by SomaScan (with clustering information).

Supplementary Table 2. Longitudinal fold-change (FC) profile and associated p-values from the Ebola patient's samples relative to day 7.

Supplementary Table 3. Longitudinal fold-change (FC) profile and associated p-values from the Ebola patient's samples relative to the mean protein abundance observed in identically irradiated pooled healthy control samples.

Supplementary Table 4. List of individual proteins selected for group analyses with functional annotations.

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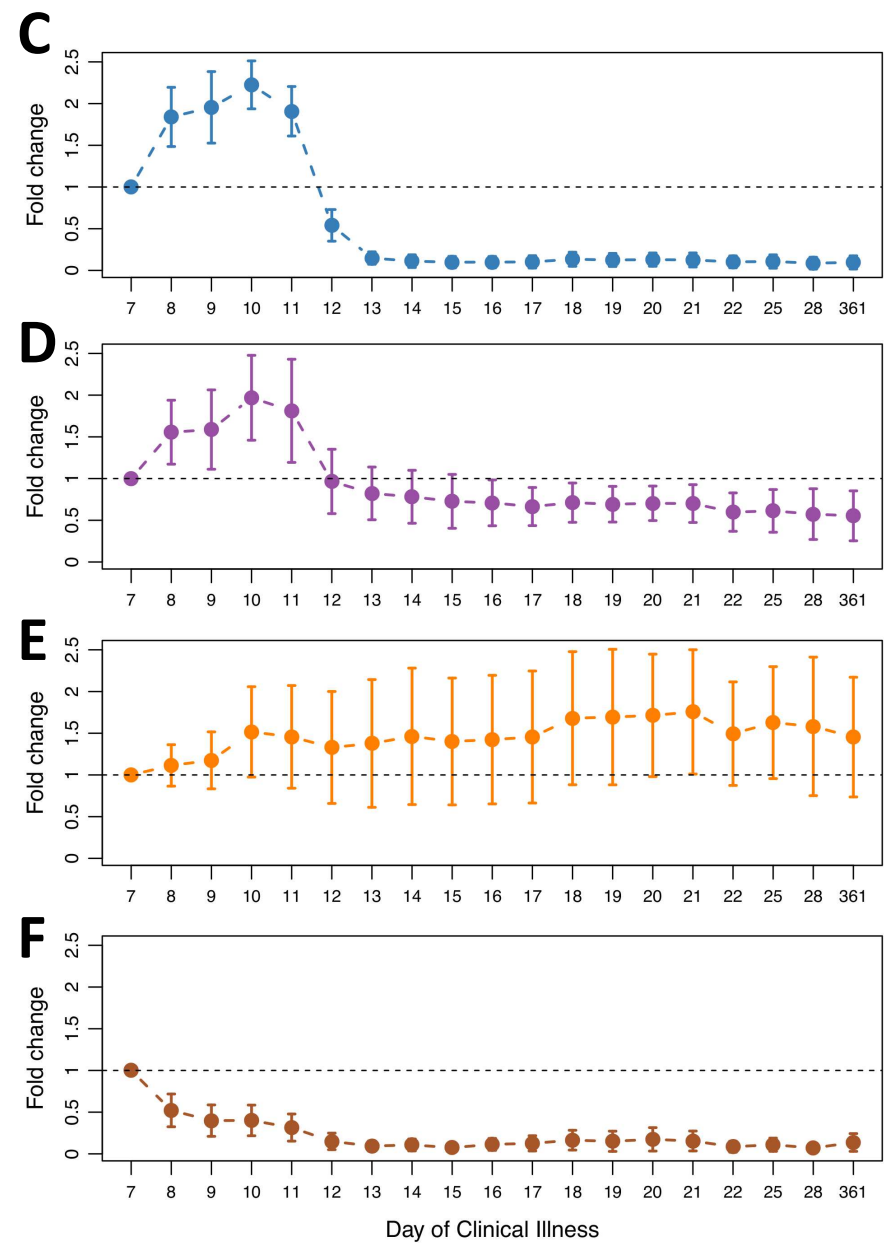
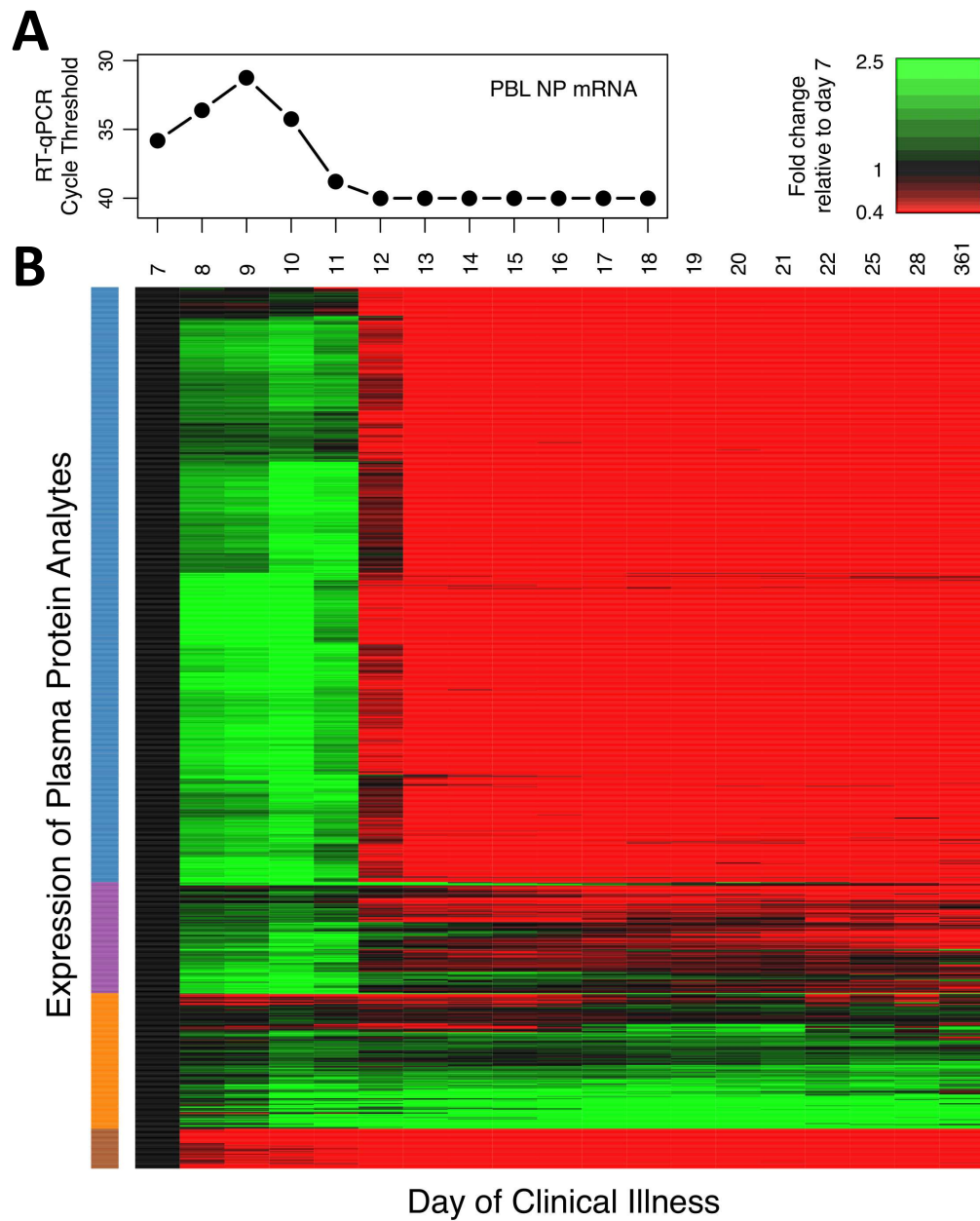


Figure 1

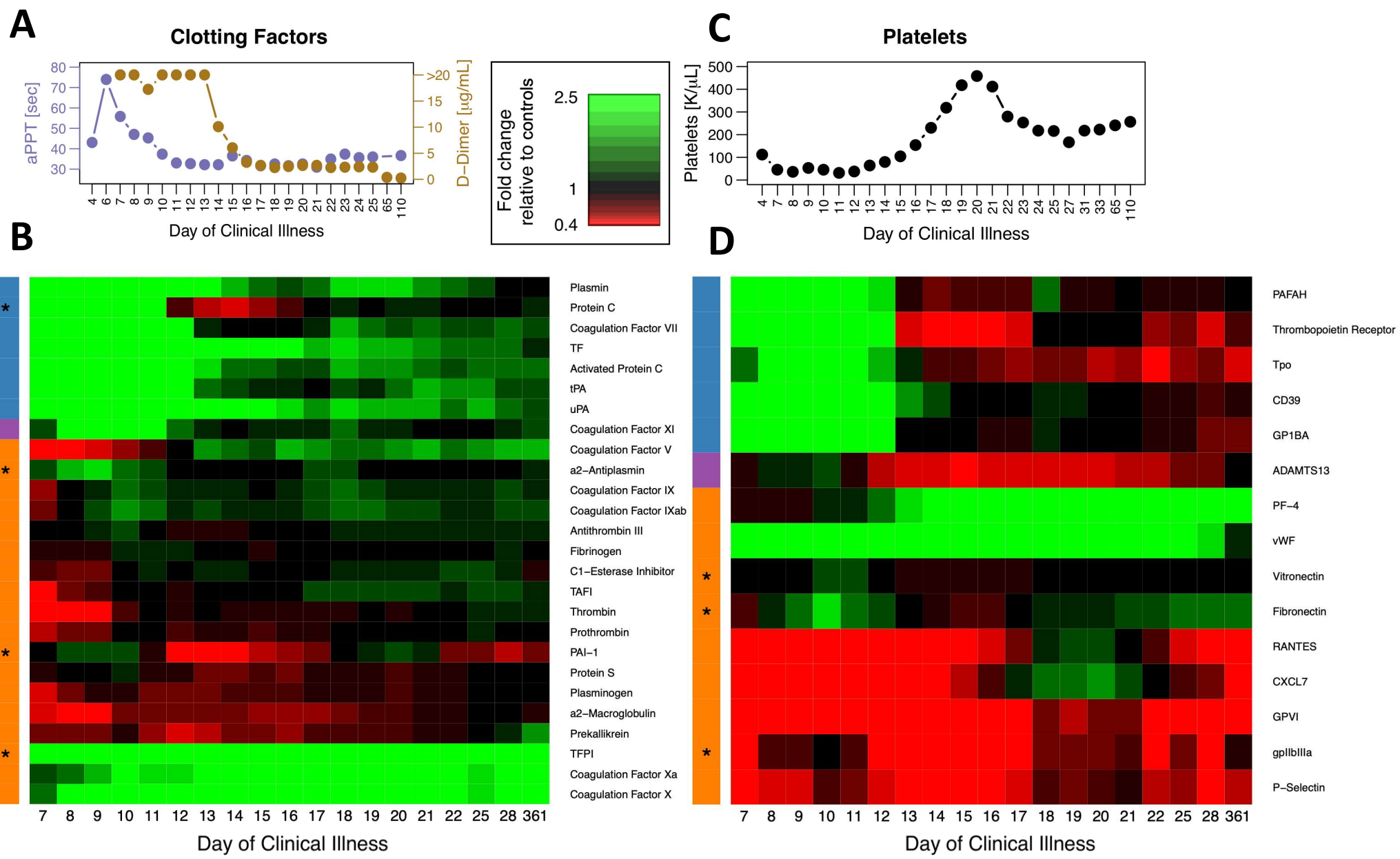


Figure 2

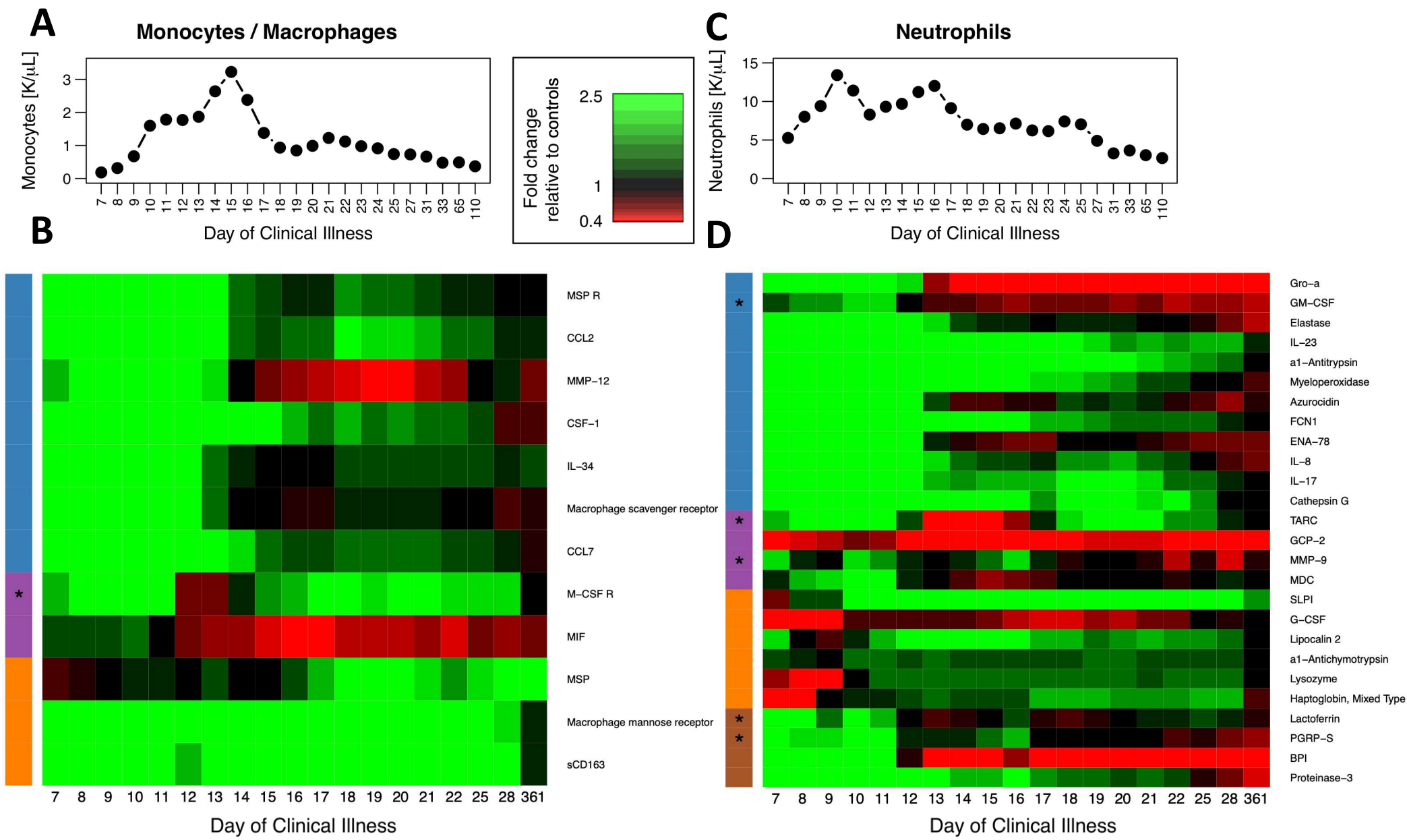


Figure 3

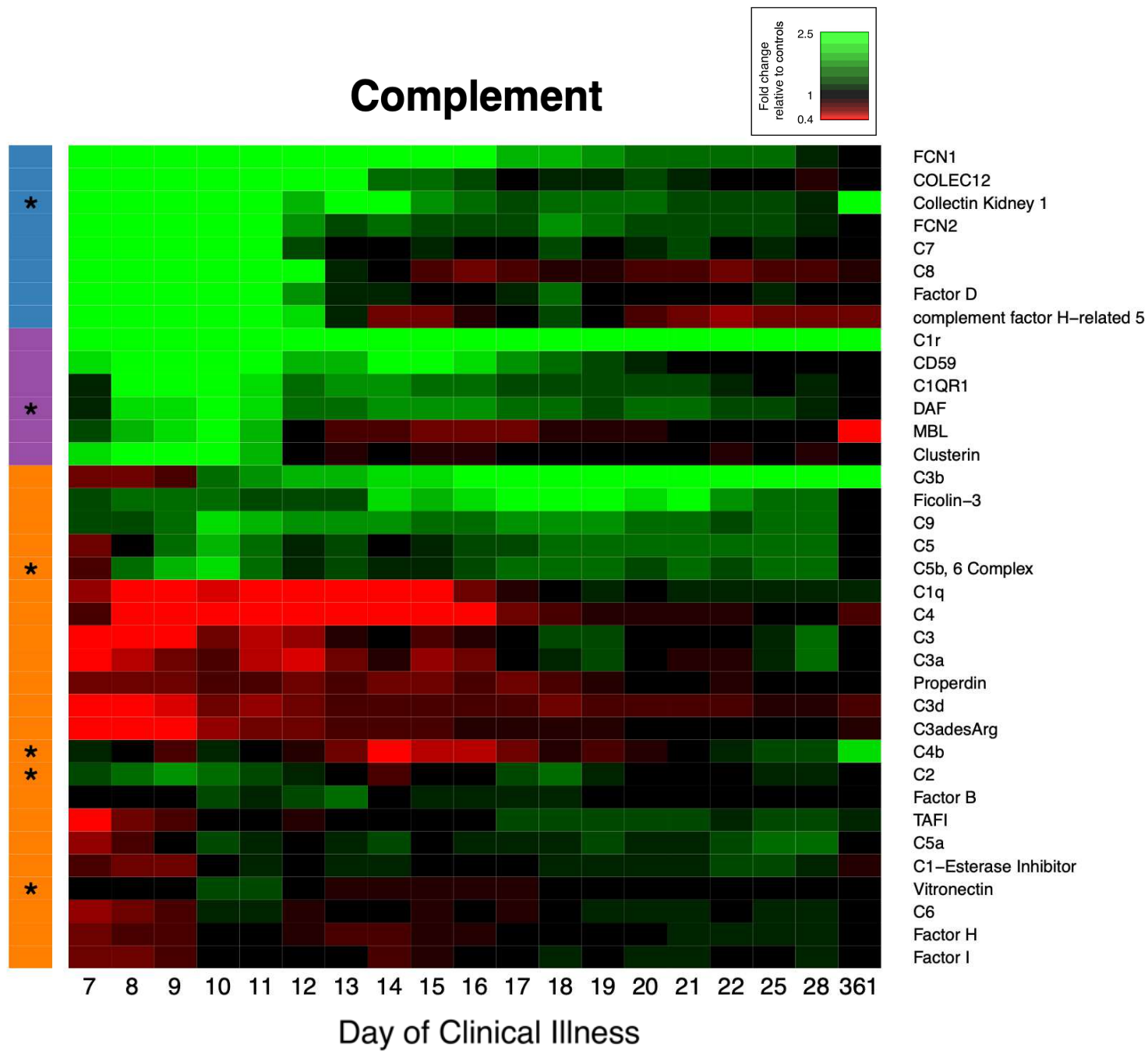


Figure 4

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